

# Water Quality Classification via Cost-Efficient Machine Learning: A Case Study in Nuevo León

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**Abstract.** Accurate water-quality assessment is vital, but laboratory costs limit monitoring in many regions. We test whether a small, low-cost indicator panel can classify Water Quality Index (WQI) categories in Nuevo León, Mexico. Using 1,302 REMANECA samples, we computed WQI with a weighted multiplicative model and trained five classifiers (RF, SVM, DT, KNN, NB) on physicochemical features. Cross-validation ranked Random Forest (RF) best with 11 indicators (accuracy  $0.921 \pm 0.023$ ; weighted F1  $0.912 \pm 0.028$ ; macro precision  $0.926 \pm 0.037$ ; macro recall  $0.785 \pm 0.073$ ). Feature selection and importances emphasized total hardness, coliforms, nutrients ( $\text{PO}_4$ ,  $\text{NO}_3^-$ ,  $\text{NH}_3$ ), and pH. A cost-aware five-test panel (hardness,  $\text{PO}_4$ , pH,  $\text{NH}_3$ , SST) retained strong performance (RF accuracy  $0.857 \pm 0.026$ ; weighted F1  $0.829 \pm 0.030$ ) with reduced minority-class sensitivity (macro recall  $0.615 \pm 0.059$ ). Errors concentrated between adjacent categories; detection of heavily contaminated water remained stable (recall 98% to 97%) and the majority class stayed high (99% to 98%), while “excellent” and “slightly contaminated” degraded. These results show that reliable WQI classification is achievable with a compact, low-cost indicator set. A tiered strategy—screen with the five-test panel and confirm with the full suite—can expand coverage under fixed budgets while preserving identification of severe contamination.

**Keywords:** Water Quality Index (WQI) · Water Quality Assessment · Machine Learning · Environmental Data Science · Mexico

## 1 Introduction

Freshwater supports ecosystems, public health, and economies, yet its availability is increasingly threatened by overuse, climate change, and diffuse pollution [13]. In response, reliable scientific indicators are essential for protecting aquatic environments and ensuring equitable access to clean water [9]. Water quality indices (WQIs) condense complex monitoring data into a single, interpretable score, enabling rapid assessments of status and trends [7]. Among the earliest and most

enduring models, the Dinius multiplicative index [6] balances sensitivity and clarity. However, human activities—ranging from nutrient-rich agricultural runoff to inadequately treated urban discharge—continue to degrade water quality. These stressors disrupt biogeochemical cycles, drive eutrophication, and weaken the natural self-purification of rivers and aquifers [3]. Prolonged exposure to contaminants such as fluoride, nitrate, and nitrite is associated with health risks, including skeletal fluorosis and methemoglobinemia [13].

To address regional challenges, several adaptations of the classic WQI have emerged. Notable examples include the Canadian Council of Ministers of the Environment (CCME) formulation; the root-mean-square WQI (RMS-WQI) and its weighted quadratic variant; and Mexico’s *Índice de Calidad del Agua*. These approaches improve robustness against outliers and redundancy [18]. Alongside these innovations, supervised machine learning (ML) has transformed water quality assessment. International case studies—spanning Bangladesh [17], China [20], Iran [14], Turkey [19], India [15], and Mexico [10]—show how data-driven models adapt across varied climates and geologies. Classical learners—Support Vector Machines (SVM), Random Forests (RF), and gradient-boosted trees—perform well when training data are plentiful and clean, classifying water quality from physicochemical inputs [11]. More recently, deep models—such as CNN-LSTM hybrids and stacked ensemble LSTMs—report accuracies above 95%, especially when paired with advanced feature-engineering methods, including variational mode decomposition and metaheuristic optimization [9].

Despite these advances, key limitations persist. In Mexico, monitoring is constrained by sparse spatial and temporal coverage [4]; feature-level sparsity (many physicochemical variables have few observations per station); and the computational cost of remote-sensing data. Regulatory standards, such as Mexico’s NOM-127-SSA1-2021 [16], set thresholds for hundreds of parameters, underscoring the need for real-time systems that translate sensor data into compliance alerts. Yet in many countries, measuring the full parameter set is cost-prohibitive and reduces data availability. A practical alternative is to train ML models that achieve comparable performance using only inexpensive variables.

In this study, we develop an ML model that classifies water quality using low-cost physicochemical indicators. We hypothesize that a reduced-feature model will exceed a random baseline (accuracy  $> 50\%$ ), and we compare its performance with a full-feature model. We evaluate five supervised algorithms using a WQI target derived from REMANECA data in Nuevo León, Mexico.

The article is organized as follows. Section 2 describes data preprocessing, WQI calculation, and model development; Section 3 reports results for full and reduced-feature models; Section 4 discusses implications for cost-effective water monitoring in resource-constrained regions.

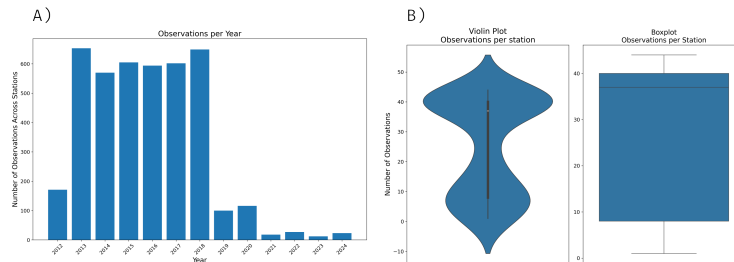
## 2 Methodology

We develop a machine-learning model to classify water quality in Nuevo León, Mexico. The pipeline uses the REMANECA dataset, describes its structure

and preprocessing, and constructs a weighted multiplicative water quality index (WQI). We then perform feature selection, train models, and evaluate five classifiers: Random Forests (RF), Support Vector Machines (SVM), Decision Trees (DT), k-nearest neighbors (KNN), and Naive Bayes (NB). Performance is measured by accuracy, precision, and F1 score, with attention to the trade-off between predictive accuracy and feature (measurement) cost. Code and materials are available at <https://github.com/kaledcorona/water-quality-classification-paper>.

## 2.1 Data

The *Comisión Nacional del Agua* (CONAGUA) is the Mexican government agency responsible for managing, regulating, and protecting national water resources [1]. Among its initiatives is the *Red Nacional de Medición de Calidad del Agua* (REMANECA) [2], a nationwide network that monitors water quality through distributed sampling stations. We use REMANECA records from Nuevo León collected during 2012 to 2024 across 163 stations: 105 lotic (rivers/streams), 40 subterranean (groundwater), 10 lentic (lakes/reservoirs), and 8 wetlands (4 lotic, 4 lentic). In total, the subset contains 4,140 samples. Sampling is uneven across stations (See Figure 1 B): the mean is about 25 samples per station (min 1, max 44). Monitoring frequency is irregular across years and not strictly monthly; 2013 to 2016 show consistent high-frequency sampling, whereas 2012 and 2021 to 2024 have fewer observations (Figure 1 A). The most sampled years are 2013 (653 observations), 2018 (649), and 2015 (605). Each sample reports up to 192 physicochemical variables (e.g., heavy metals, dissolved oxygen, microbial and chemical contaminants), plus six features from station’s metadata. To reduce sparsity, we excluded 50 stations with fewer than 10 observations from model training and evaluation.



**Fig. 1.** Sampling intensity in REMANECA (Nuevo León). (A) Annual observations across 163 stations, 2012–2024—high in 2013–2018, sparse after 2019. (B) Observations per station (violin and boxplot) show wide heterogeneity (1–44 samples per station).

## 2.2 Data Preprocessing

We preprocess the data in Python 3.12.11 using NumPy (version 2.0.2), pandas (2.2.2), and scikit-learn (1.6.1). Initial inspection showed type issues: numeric fields stored as pandas `object` (strings) and station metadata lacking categorical dtypes. Some measurements also appeared as string ranges (e.g., “<4”, “>200”). To correct these, Algorithm 1 removes non-numeric symbols, coerces values to numeric, and standardizes missing entries.

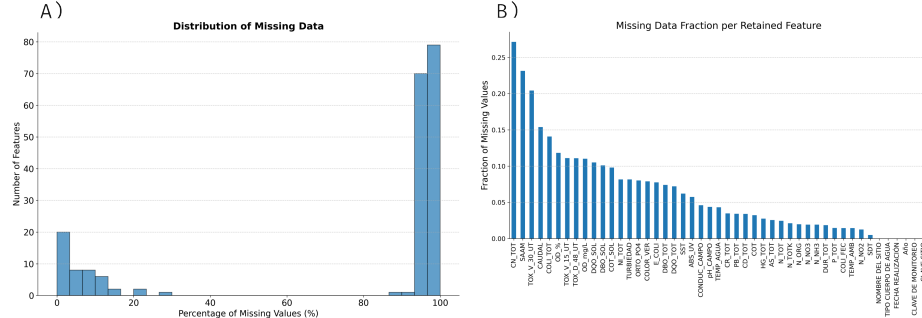
We then filtered features with more than 50% missing values. A histogram of feature-wise missingness (Figure 2 A) showed a clear gap between 35% and 85%, making 50% a natural cutoff. In total, 151 features were removed, yielding a reduced set  $X' \subset X$  with 47 features for downstream processing (Figure 2 B).

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**Algorithm 1** String Cleaning and Type Coercion for Numerical Features

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- 1: **for each** feature `col` in  $X$  **do**
  - 2:   Transform values in  $x_n \subset X$  to text,
  - 3:   Replace commas (‘,’) by dots (‘.’) in  $x_n$ ,
  - 4:   Delete ‘<’ or ‘>’ in  $x_n$ ,
  - 5:   Eliminate letters, spaces, and symbols in  $x_n$ ,
  - 6:   Replace string missing values by NaN
  - 7:   Transform  $x_n$  to numeric values; errors are converted to NaN
  - 8:   Assign  $x_n$  back to  $X$ .
  - 9: **end for**
- 



**Fig. 2.** Feature-wise missingness in REMANECA (Nuevo León). (A) Histogram of the percentage of missing values across 192 variables shows a clear split between low-missingness features and highly sparse ones, motivating a 50% exclusion threshold. (B) Missing-data fractions for the 47 retained features (sorted), the majority with < 15% missingness.

Although filtering improved completeness, many variables in  $X'$  still contained missing values. We applied hierarchical median imputation—first within

station-year, then within station, and finally globally. Overall, 6% of values were imputed. The most imputed features were total cyanides (27%), methylene blue active substances (23%), and *Vibrio fischeri* toxicity (30 min; 20%).

We then addressed outliers using interquartile-range rules. Mild outliers were values outside  $[Q_1 - 1.5 \text{ IQR}, Q_3 + 1.5 \text{ IQR}]$ ; extreme outliers were outside  $[Q_1 - 3 \text{ IQR}, Q_3 + 3 \text{ IQR}]$ . We removed extreme outliers and retained mild ones, which comprised a sizable fraction of the data. To limit temporal sparsity, we removed 44 samples from 2021–2024, which were insufficient for reliable generalization. We also excluded 52 underrepresented stations with fewer than three observations. The final dataset contains 1,302 samples and 47 features.

After imputation and outlier filtering, we selected variables according to Dinius [6], restricted to those available in  $X'$ . The original scheme specifies 15 parameters; our dataset contained 11 of these, with Dinius weights  $W_i$ : dissolved oxygen (DO) 0.103, biochemical oxygen demand (BOD) 0.096, chemical oxygen demand (COD) 0.053, pH 0.063, total suspended solids (TSS) 0.033, total coliforms (coli fec) 0.083, fecal coliforms (fecal coli) 0.143, nitrate nitrogen ( $\text{NO}_3^-$ -N) 0.053, ammoniacal nitrogen ( $\text{NH}_3$ -N) 0.043, orthophosphates ( $\text{PO}_4^{3-}$ -P) 0.073, and total hardness ( $\text{CaCO}_3$ ) 0.058. The resulting set  $H \subset X'$  comprises 11 features used in subsequent analyses. Using  $H$ , we computed the WQI with a weighted multiplicative model.

### 2.3 Water Quality Index

The water quality index (WQI) summarizes physicochemical and biological indicators on a 0–100 scale (higher is better). Following Brown et al. (1973) [5] and its adaptation for Mexico by Dinius [6], each measurement  $x_i$  is mapped to a subindex  $Q_i \in [0, 100]$  using rating tables/curves based on expert judgment reported in the literature. We compute  $Q_i$  from Dinius’s tables (Table 1) via piecewise-linear interpolation between tabulated pairs  $(x_i, Q_i)$ . Parameters are assigned weights  $W_i \in [0, 1]$  to reflect their relative importance. The overall index is the weighted multiplicative function; for numerical stability we use its logarithmic form (Eq. 1).

Because only 11 of the original 15 parameters were available, we used the corresponding Dinius weights and renormalized them so that  $\sum_{i=1}^n W_i = 1$  (with  $W_i \geq 0$ ,  $n = |H| = 11$ ). The WQI is the weighted geometric mean of the subindices:

$$\text{WQI} = \exp\left(\sum_{i=1}^n W_i \log Q_i\right), \quad Q_i \in (0, 100]. \quad (1)$$

For agricultural use, WQI values are classified as: excellent ( $E$ , 90–100), acceptable ( $A$ , 70–90), slightly contaminated ( $LC$ , 50–70), contaminated ( $C$ , 30–50), heavily contaminated ( $FC$ , 20–30), and extremely contaminated ( $EC$ , 0–20). These categories were mapped to ordinal labels for learning:  $EC=0$ ,  $FC=1$ ,  $C=2$ ,  $LC=3$ ,  $A=4$ ,  $E=5$ . The next section describes the models used to predict these classes from  $H$ .

**Table 1.** Rating curves used to compute subindices  $Q_i$  (linear interpolation between breakpoints).

| Parameter | Curve points $(x, Q)$                                                                                                    |
|-----------|--------------------------------------------------------------------------------------------------------------------------|
| OD_%      | (20, 10), (40, 25), (60, 50), (80, 75), (90, 90), (100, 100), (120, 90), (140, 50)                                       |
| DBO_SOL   | (30, 0), (15, 25), (8, 50), (5, 75), (3, 100)                                                                            |
| DQO_SOL   | (300, 0), (140, 25), (70, 50), (40, 75), (20, 100)                                                                       |
| SST       | (400, 0), (150, 25), (75, 50), (40, 75), (20, 100)                                                                       |
| COLI_TOT  | (100000, 0), (20000, 25), (5000, 50), (1000, 60), (200, 70), (0, 100)                                                    |
| COLI_FEC  | (100000, 0), (20000, 25), (5000, 50), (1000, 60), (200, 70), (0, 100)                                                    |
| N_NO3     | (100, 2), (40, 10), (15, 25), (10, 70), (7, 80), (3, 90), (1, 100)                                                       |
| N_NH3     | (5.0, 0), (2.0, 25), (1.0, 50), (0.5, 75), (0.1, 100)                                                                    |
| ORTO_PO4  | (1.00, 0), (0.40, 25), (0.20, 50), (0.10, 70), (0.05, 80), (0.00, 100)                                                   |
| pH_CAMPO  | (2.0, 0), (5.0, 25), (6.0, 50), (6.5, 75), (7.2, 95), (7.4, 100), (7.6, 95), (8.5, 75), (9.0, 50), (10.0, 25), (12.0, 0) |
| DUR_TOT   | (0, 0), (20, 25), (80, 100), (120, 100), (200, 75), (300, 50), (450, 25), (600, 0)                                       |

## 2.4 Model Description

We evaluated five scikit-learn classifiers (defaults unless noted): Random Forest (RF; `class_weight=balanced`), Support Vector Classifier (SVC),  $k$ -nearest neighbors ( $k = 5$ ; KNN), Gaussian Naive Bayes (NB), and Decision Tree (DT). A fixed random seed (42) controlled the split and model randomness. Data were partitioned with a stratified 80/20 train–test split (1,041/261 samples). Features were standardized (z-scores) on the training set and the transformation applied to the test set; trees do not require scaling, but we used a single pipeline for consistency. Within the training set, we ran stratified 10-fold cross-validation to estimate variability, refit on the full training data, and reported results on the held-out test set. Metrics were accuracy plus macro- and weighted-averaged precision, recall, and F1.

As a continuous baseline, we fit ordinary least squares to predict WQI from  $H$  (normalized), reporting mean squared error (MSE) and adjusted  $R^2$ . We inspected standardized coefficients and RF-based importances, and applied recursive feature elimination with cross-validation (RFECV) using the WQI class target; the RMSE plateaued at five features.

Guided by this result and test-pricing data [8], we trained a simplified classifier using five lower-price variables—pH, SST,  $\text{NH}_3$ ,  $\text{PO}_4$ , and total hardness—following the same protocol. We empirically chose these features to be part of the set of important and be of low cost.

**Table 2.** Cross-validated performance (mean  $\pm$  SD across folds).

|                    | DT                | KNN               | NB                | RF                | SVM               |
|--------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| accuracy           | 0.898 $\pm$ 0.020 | 0.822 $\pm$ 0.029 | 0.793 $\pm$ 0.039 | 0.921 $\pm$ 0.023 | 0.866 $\pm$ 0.019 |
| precision macro    | 0.829 $\pm$ 0.049 | 0.714 $\pm$ 0.077 | 0.682 $\pm$ 0.060 | 0.926 $\pm$ 0.037 | 0.653 $\pm$ 0.037 |
| recall macro       | 0.830 $\pm$ 0.036 | 0.604 $\pm$ 0.059 | 0.815 $\pm$ 0.063 | 0.785 $\pm$ 0.073 | 0.634 $\pm$ 0.045 |
| f1 macro           | 0.823 $\pm$ 0.031 | 0.641 $\pm$ 0.067 | 0.719 $\pm$ 0.060 | 0.831 $\pm$ 0.061 | 0.635 $\pm$ 0.043 |
| precision weighted | 0.904 $\pm$ 0.018 | 0.809 $\pm$ 0.039 | 0.859 $\pm$ 0.032 | 0.923 $\pm$ 0.022 | 0.810 $\pm$ 0.020 |
| recall weighted    | 0.898 $\pm$ 0.020 | 0.822 $\pm$ 0.029 | 0.793 $\pm$ 0.039 | 0.921 $\pm$ 0.023 | 0.866 $\pm$ 0.019 |
| f1 weighted        | 0.898 $\pm$ 0.019 | 0.808 $\pm$ 0.034 | 0.808 $\pm$ 0.036 | 0.912 $\pm$ 0.028 | 0.833 $\pm$ 0.022 |

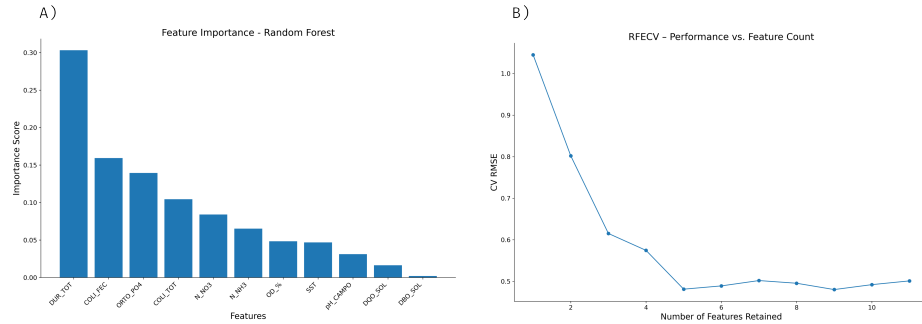
DT: Decision Tree; k-NN: K-Nearest Neighbors; NB: Naïve Bayes; RF: Random Forest; SVM: Support Vector Machine.

### 3 Results

*Full (11-feature) models.* Cross-validated results (Table 2) rank Random Forest (RF) best overall (accuracy  $0.921 \pm 0.023$ , weighted F1  $0.912 \pm 0.028$ ) and with the highest macro precision ( $0.926 \pm 0.037$ ). Its macro recall is lower ( $0.785 \pm 0.073$ ), indicating reduced sensitivity to minority classes. Decision Tree (DT) is competitive on accuracy ( $0.898 \pm 0.020$ ) and attains the best class balance (macro recall  $0.830 \pm 0.036$ ). SVM and KNN are mid-range (accuracies  $0.866 \pm 0.019$  and  $0.822 \pm 0.029$ ). Naïve Bayes (NB) has the lowest accuracy ( $0.793 \pm 0.039$ ) but relatively strong macro recall ( $0.815 \pm 0.063$ ), reflecting recall-biased predictions.

As a continuous baseline, ordinary least squares on  $H$  achieved adjusted  $R^2 = 0.826$  and RMSE = 113.15 (WQI units). Coefficient magnitudes highlighted orthophosphate ( $\text{PO}_4$ ; negative), biochemical oxygen demand (DBO; positive), pH (negative), and ammonia ( $\text{NH}_3$ ; positive) as the most influential predictors. RF feature importance (Figure 3) similarly emphasized total hardness, fecal coliforms,  $\text{PO}_4$ , total coliforms, nitrate ( $\text{N\_NO}_3$ ), and ammonia ( $\text{N\_NH}_3$ ). RFECV (10-fold) with categories of WQI target showed steep gains up to five features and diminishing returns thereafter; the lowest CV-RMSE occurred near nine features, indicating that most predictive signal is captured by a compact subset.

*Five-feature models.* Using five low-cost variables (DUR\_TOT,  $\text{PO}_4$ , pH,  $\text{NH}_3$ , SST), performance declines moderately but remains strong (Table 3). RF accuracy drops by 6.4 percentage points (to  $0.857 \pm 0.026$ ); weighted F1 decreases by 8.3 points (to  $0.829 \pm 0.030$ ). SVM is similar ( $0.854 \pm 0.016$ ), KNN lower ( $0.838 \pm 0.029$ ), and DT/NB trail.



**Fig. 3.** Feature selection summary. (A) Random Forest feature importance highlights hardness (DUR\_TOT) and coliforms as top contributors. (B) RFECV curve (CV RMSE vs. features retained) improves steeply up to  $\sim 5$  features and then plateaus, supporting a compact panel.

*Error patterns.* Confusion matrices for RF (Fig. 4) make the trade-off explicit. With 11 features, errors concentrate in adjacent classes— $LC \rightarrow A$  (42),  $E \rightarrow A$  (27), and  $EC \rightarrow FC$  (16). Per-class accuracies are A 768/776 (99%), LC 50/93 (54%), FC 267/272 (98%), E 54/81 (67%), and EC 60/80 (75%). Using five features increases these confusions:  $LC \rightarrow A$  rises to 56,  $E \rightarrow A$  to 74,  $A \rightarrow E$  to 15, and  $EC \rightarrow FC$  to 26 ( $FC \rightarrow EC$  9). FC remains largely preserved (267  $\rightarrow$  263 correct) and A stays high (768  $\rightarrow$  757), but E degrades sharply (54/81  $\rightarrow$  7/81).

## 4 Discussion

This study shows that WQI classes can be predicted reliably with standard models and a compact set of indicators. The full 11-feature configuration delivers the strongest overall performance; the five-feature panel trades some minority-class sensitivity for substantial savings in measurement cost. Errors concentrate between adjacent categories (A–E, A–LC), while detection of heavily contaminated water remains stable—an attractive property for screening.

Model behaviors align with domain knowledge. Hardness and coliforms dominate, and nutrients ( $PO_4$ ,  $NO_3^-$ ,  $NH_3$ ) and pH carry substantial signal, consistent with the literature and the continuous baseline. Feature-elimination curves suggest quickly diminishing returns beyond a small subset, reinforcing the case for cost-aware monitoring.

These results address a different task than Molina et al. [12], who trained a Decision Tree to classify water use types—drinking, recreation, irrigation, and aquatic life—achieving 99% accuracy with 45 features. Direct accuracy comparison is not meaningful. Our focus is WQI-category prediction with an emphasis on operational affordability.

*Practical implications.* A tiered strategy is feasible. Use the five-test panel (hardness,  $PO_4$ , pH,  $NH_3$ , SST) for routine sampling, and trigger the full suite only

**Table 3.** Cross-validated performance with five-feature models (mean  $\pm$  SD across folds).

|                    | DT                | KNN               | NB                | RF                | SVM               |
|--------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| accuracy           | 0.783 $\pm$ 0.028 | 0.838 $\pm$ 0.029 | 0.682 $\pm$ 0.034 | 0.857 $\pm$ 0.026 | 0.854 $\pm$ 0.016 |
| precision macro    | 0.613 $\pm$ 0.054 | 0.727 $\pm$ 0.111 | 0.676 $\pm$ 0.046 | 0.758 $\pm$ 0.084 | 0.693 $\pm$ 0.054 |
| recall macro       | 0.608 $\pm$ 0.052 | 0.587 $\pm$ 0.051 | 0.702 $\pm$ 0.044 | 0.615 $\pm$ 0.059 | 0.581 $\pm$ 0.029 |
| f1 macro           | 0.607 $\pm$ 0.050 | 0.619 $\pm$ 0.060 | 0.633 $\pm$ 0.033 | 0.647 $\pm$ 0.066 | 0.599 $\pm$ 0.033 |
| precision weighted | 0.783 $\pm$ 0.028 | 0.816 $\pm$ 0.045 | 0.818 $\pm$ 0.027 | 0.833 $\pm$ 0.035 | 0.804 $\pm$ 0.021 |
| recall weighted    | 0.783 $\pm$ 0.028 | 0.838 $\pm$ 0.029 | 0.682 $\pm$ 0.034 | 0.857 $\pm$ 0.026 | 0.854 $\pm$ 0.016 |
| f1 weighted        | 0.782 $\pm$ 0.027 | 0.812 $\pm$ 0.031 | 0.715 $\pm$ 0.029 | 0.829 $\pm$ 0.030 | 0.812 $\pm$ 0.018 |

DT: Decision Tree; KNN:  $k$ -Nearest Neighbors; NB: Naïve Bayes; RF: Random Forest; SVM: Support Vector Machine.

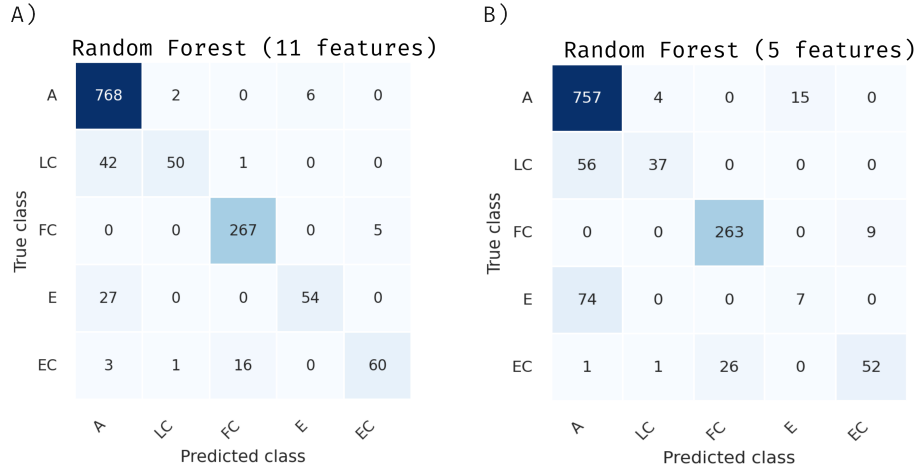
when the model flags uncertainty or potential noncompliance. This approach can expand spatial coverage or sampling frequency under fixed budgets while preserving reliable identification of severe contamination.

*Limitations.* (i) Single-region data (Nuevo León) may limit generalization. (ii) The “contaminated” class was underrepresented after WQI computation, affecting class balance. (iii) Features with  $> 50\%$  missingness and sparse years/stations were removed; informative signals may have been lost. (iv) Type coercion, imputation, rating-curve choices, and weight renormalization introduce assumptions. (v) Irregular sampling across years may bias temporal patterns. (vi) Cost figures rely on secondary sources; primary pricing is needed for firm economic claims.

*Future work.* Validate across basins; adopt cost-sensitive and ordinal objectives; use temporal holdouts to test robustness; and explore alternative or locally calibrated WQI formulations. Integrating auxiliary data (e.g., flow, land use, remote sensing) may improve recall for minority classes without substantially increasing cost.

## 5 Conclusions

Using 1,302 samples from REMANECA, we built WQI-based classifiers for Nuevo León and compared a full 11-indicator model with a cost-aware five-indicator panel. With all 11 features, Random Forest (RF) achieved  $0.921 \pm 0.023$  accuracy and  $0.912 \pm 0.028$  weighted F1; with five features (hardness,  $\text{PO}_4$ , pH,



**Fig. 4.** Confusion matrices for Random Forest models. (A) Full model with 11 features; (B) reduced model with five features.

NH<sub>3</sub>, SST), RF reached  $0.857 \pm 0.026$  and  $0.829 \pm 0.030$ , respectively. This corresponds to declines of 6.4 and 8.3 percentage points, and a macro-recall drop from  $0.785 \pm 0.073$  to  $0.615 \pm 0.059$  (17 points). Despite this, per-class recall for the screening-critical “heavily contaminated” level remained high ( $98\% \rightarrow 97\%$ ), and the majority class stayed robust (A:  $99\% \rightarrow 98\%$ ), while “excellent” degraded ( $67\% \rightarrow 9\%$ ) and “slightly contaminated” fell ( $54\% \rightarrow 40\%$ ).

Operationally, the five-test panel reduces assays by 55% per site ( $11 \rightarrow 5$ ) while retaining 0.85 accuracy with RF/SVM, enabling broader spatial or temporal coverage under fixed budgets. The full 11-feature model offers better minority-class sensitivity and is suited for confirmatory testing.

Limitations include single-region data, irregular sampling, and class sparsity; external validation, cost-sensitive and ordinal objectives, and temporal holdouts are natural next steps. All code, rating curves, and weights are openly released to support reuse and extension.

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**Disclosure of Interests.** The authors declare no competing interests.

## References

1. del Agua, C.N.: Acerca del organismo [about the organism] (2025), <https://www.gob.mx/conagua/acciones-y-programas/acerca-del-organismo>

2. del Agua, C.N.: Resultados de la red nacional de medición de calidad del agua (renameca) [results of the national water quality monitoring network] (2025), <https://www.gob.mx/conagua/es/articulos/resultados-de-la-red-nacional-de-medicion-de-calidad-del-agua-renameca?idiom=es>, source of the data set used in the article.
3. Akhtar, N., Syakir Ishak, M.I., Bhawani, S.A., Umar, K.: Various natural and anthropogenic factors responsible for water quality degradation: A review. *Water* **13**(19), 2660 (2021)
4. AriasRodriguez, L.F., Duan, Z., Sepúlveda, R., MartinezMartinez, S.I., Disse, M.: Monitoring water quality of valle de bravo reservoir, mexico, using entire lifespan of meris data and machine learning approaches. *Remote Sensing* **12**(10), 1586 (2020)
5. Brown, R., McClelland, N., Deininger, R., Landwehr, J.: Validating the wqi. In: *Proceedings of the National Meeting of American Society of Civil Engineers on Water Resources Engineering*, Washington, DC, USA. vol. 29 (1973)
6. Dinius, S.: Design of an index of water quality 1. *JAWRA Journal of the American Water Resources Association* **23**(5), 833–843 (1987)
7. Gitau, M.W., Chen, J., Ma, Z.: Water quality indices as tools for decision making and management. *Water resources management* **30**, 2591–2610 (2016)
8. Institute, C.S.: Non-potable water tests & fees (2025), <https://communityscience.org/certified-water-testing/surface-water-tests-fees/>, accessed: June 6, 2025
9. Karpagam, V., Christy, S., et al.: Deep learning-based water quality index classification using stacked ensemble variational mode decomposition. *Environmental Research Communications* **6**(6), 065005 (2024)
10. López, E., Patiño, R., Vázquez-Sauceda, M.L., Pérez-Castañeda, R., Arellano-Méndez, L.U., Ventura, R., Heyer, L.: Water quality and ecological risk assessment of intermittent streamflow through mining and urban areas of san marcos river sub-basin, mexico. *Environmental Nanotechnology, Monitoring & Management* **14**, 100369 (2020)
11. Mohammadpour, A., Gharehchahi, E., Golaki, M., Gharaghani, M.A., Ahmadian, F., Abolfathi, S., Samaei, M.R., Uddin, M.G., Olbert, A.I., Khaneghah, A.M.: Advanced water quality assessment using machine learning: Source identification and probabilistic health risk analysis. *Results in Engineering* p. 105421 (2025)
12. Molina, Y., López, J.V.P., Castro, H.M.C., Gonzalez-Mendoza, M.: Predicting and classifying contaminants in mexican water bodies: A machine learning approach. In: *Mexican International Conference on Artificial Intelligence*. pp. 40–53. Springer (2024)
13. Moloantoa, K.M., Khetsha, Z.P., Van Heerden, E., Castillo, J.C., Cason, E.D.: Nitrate water contamination from industrial activities and complete denitrification as a remediation option. *Water* **14**(5), 799 (2022)
14. Noori, R., Farahani, F., Jun, C., Aradpour, S., Bateni, S.M., Ghazban, F., Hosseinzadeh, M., Maghrebi, M., Naseh, M.R.V., Abolfathi, S.: A non-threshold model to estimate carcinogenic risk of nitrate-nitrite in drinking water. *Journal of Cleaner Production* **363**, 132432 (2022)
15. Ram, A., Tiwari, S., Pandey, H., Chaurasia, A.K., Singh, S., Singh, Y.: Ground-water quality assessment using water quality index (wqi) under gis framework. *Applied Water Science* **11**, 1–20 (2021)
16. de Salud, S.: Nom-127-ssa1-2021: Agua para uso y consumo humano. límites permisibles de la calidad del agua.

- [https://www.dof.gob.mx/nota\\_detalle.php?codigo=5634136&fecha=22/03/2022](https://www.dof.gob.mx/nota_detalle.php?codigo=5634136&fecha=22/03/2022) (2021), diario Oficial de la Federación
17. Uddin, M.G., Diganta, M.T.M., Sajib, A.M., Hasan, M.A., Moniruzzaman, M., Rahman, A., Olbert, A.I., Moniruzzaman, M.: Assessment of hydrogeochemistry in groundwater using water quality index model and indices approaches. *Heliyon* **9**(9) (2023)
  18. Uddin, M.G., Nash, S., Olbert, A.I.: A review of water quality index models and their use for assessing surface water quality. *Ecological Indicators* **122**, 107218 (2021)
  19. Ustaoglu, F., Tepe, Y., Tas, B.: Assessment of stream quality and health risk in a subtropical turkey river system: A combined approach using statistical analysis and water quality index. *Ecological indicators* **113**, 105815 (2020)
  20. Zhang, Q., Xu, P., Qian, H.: Groundwater quality assessment using improved water quality index (wqi) and human health risk (hhr) evaluation in a semi-arid region of northwest china. *Exposure and health* **12**(3), 487–500 (2020)